

Mourning Skins: Using Generative AI for Speculative Visualization on Climate Change

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Figure 1: Overview of Mourning Skins, presenting speculative visualizations of four species: (a) Polar Bear, (b) Butterfly, (c) Antelope, and (d) Coral. The left shows the rendered visualization, and the right shows the actual exhibition display.

Abstract

Mourning Skins is a generative AI-driven project that uses speculative visualization to explore how climate change may reshape the appearances of species. Grounded in multimodal datasets and AI-driven speculation, the project integrates structured prompts, diffusion-based image and video generation to envision long-term transformations of four representative species. By rendering the slow violence of climate change into visual narratives, the work reshapes the relationship between humans and nature.

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1 Introduction

Climate change imposes profound yet often imperceptible pressures on ecosystems, driving gradual but irreversible transformations in species' morphology, behavior, and habitats. Although these changes are well-documented in scientific literature, their extended temporal scales often render them beyond the reach of everyday human perception. Consequently, we employ a speculative visualization approach [Kim and DiSalvo 2010], which advocates the

critical use of visual rhetoric to stimulate public discourse around socially significant data.

Our project, *Mourning Skins*, uses generative AI to create speculative visualizations of transformations in animals' fur [Gozalo-Brizuela and Garrido-Merchán 2023], as workflow outlined in Fig. 2. Drawing upon a multimodal dataset of climate-induced morphological changes, the project engages AI-driven speculative reasoning to imagine how species might adapt to extreme climate pressures. By condensing millennia of potential transformations for four species—polar bears, corals, butterflies, and antelopes into time-lapse narratives of less than one minute, the work renders the slow violence of climate change perceptible.

2 Workflow

2.1 Data collecting

We focused on the visual exploration of how global warming influences species' appearances. To support this inquiry, we curated a multimodal dataset including four species: polar bears, corals, butterflies, and antelopes. The dataset comprised three components: (a) scientific literature and datasets describing morphological impacts of climate change; (b) image collections documenting historical changes in fur patterns; (c) news articles reflecting public attitudes toward species transformations under global warming.

2.2 Speculative thinking with AI

We conducted an AI-driven speculative design phase, using the constructed datasets as input, to allow AI to envision the morphological changes that species might undergo under long-term climate pressure. Scientific literature and image collections were analyzed

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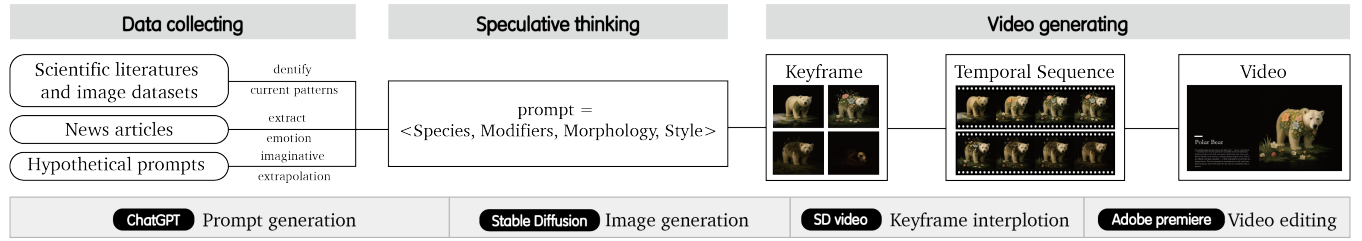


Figure 2: The workflow of Mourning Skins.



Figure 3: Keyframe examples generated by Stable diffusion.

via ChatGPT to identify transformation patterns and future trends, while news articles were mined for affective keywords such as “fragility” and “mourning,” capturing public sentiment regarding climate change.

To further inspire the imaginative extrapolation of the model, we designed hypothetical prompts framed as questions. For instance:

“Based on the uploaded data, imagine how the polar bear’s fur might transform if global warming continues to escalate toward extreme conditions.”

Based on the model’s speculative output, we then asked it to convert the imagined scenes into structured visual prompts to guide subsequent image generation. The structure of these prompts typically follows this pattern:

prompt = <Species, Modifiers, Morphology, Style>

For example, *“A polar bear, once covered in pristine white fur, now gradually overgrown with moss and wildflowers in a damp climate, its fragile body evoking a profound sense of mourning.”*

2.3 Visualization generating with AI

We generated images based on the structured prompts and constructed dynamic visual sequences from them. The prompts produced by ChatGPT were fed into stable diffusion v2.1, a diffusion-based image generation model with custom fine-tuning. The process consisted of two stages:

Keyframe Generation. Initial frames were generated via text-to-image synthesis to establish baselines for morphology and environment. Image-to-image techniques were then applied to generate

intermediate stages, maintaining consistency in structure, texture, and background. Keyframe examples are shown in Fig. 3.

Temporal Sequence Synthesis. Anchor frames were interpolated with stable video diffusion [Ho et al. 2022], combining prompt guidance and structural preservation to balance narrative coherence with local detail continuity. The resulting segments were compiled into continuous time-lapse narratives through nonlinear video editing, ensuring temporal and thematic consistency.

3 Final Presentation and Audience Perception

Mourning Skins comprises four surreal time-lapse videos, each depicting a speculative transformation under extreme climate conditions:

- **Polar Bear.** As the Arctic warms, melting ice exposes soil where plants begin to grow. In the increasingly humid climate, the bear’s once-thick white fur becomes fertile ground for moss and flowers. Over time, the fur decays and ultimately collapses.
- **Butterfly.** The intensifying heat chars forest trees. To adapt, the butterfly’s wings darken to match the carbonized habitat, before both insect and forest are consumed by fire.
- **Coral.** Rising ocean temperatures cause red coral polyps to gradually abandon the coral surface. The structure bleaches, fractures, and eventually disintegrates into dust.
- **Antelope.** As receding snow lines give way to barren rocky terrain, the antelope’s white fur slowly petrifies into brittle camouflage that erodes and breaks apart.

We presented the work in a gallery space and invited visitors to freely observe and respond. Due to equipment limitations, the videos were displayed sequentially on a single screen, as shown in Fig. 1. Many viewers described the experience as “visually striking”, noting that the gradually distorted animal forms made the abstract threat of climate change tangible.

4 Summary and Future Work

Mourning Skins demonstrates how generative AI can support speculative visualization by rendering the “slow violence” of climate change into tangible, emotionally resonant narratives. The project highlights speculative visualization’s ability to foster both affective engagement and cognitive reflection, emphasizing the potential of generative AI at the bond of art and science communication.

Looking ahead, we plan to further investigate the integration of multimodal inputs, interactive narrative branching, and adaptive rendering techniques to enhance the immersive and participatory qualities of the work.

Acknowledgments

Yechun Peng is the corresponding author.

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